```python
import streamlit as st
import pandas as pd
import json
from collections import Counter
import os

st.set_page_config(page_title="Dataset Explorer", layout="wide")

@st.cache_data
def load_data(path):
    data = []
    with open(path, "r", encoding="utf-8") as f:
        for i, line in enumerate(f):
            obj = json.loads(line)
            text = obj.get("text", "")

            for span in obj.get("spans", []):
                label = span["label"]

                if label == "OUT_OF_TOPIC":
                    continue

                try:
                    aspect, sentiment = label.split("_")
                except:
                    continue

                data.append({
                    "review_id": i,
                    "text": text,
                    "aspect": aspect,
                    "sentiment": sentiment
                })

    return pd.DataFrame(data)

BASE_DIR = os.path.dirname(__file__)
DATA_PATH = os.path.join(BASE_DIR, "..", "dataset", "Kelp3_dataset_anotasi.jsonl")
df = load_data(DATA_PATH)

st.sidebar.title("Navigasi")
menu = st.sidebar.radio("Pilih Halaman", [
    "Overview",
    "Browse Reviews",
    "Statistik",
    "Panel ABSA"
])

if menu == "Overview":
    st.title("рц"Ј Overview Dataset")

    st.write("Jumlah Data:", len(df))
    st.write("Jumlah Review:", df["review_id"].nunique())

    st.subheader("Distribusi Aspek")
    st.bar_chart(df["aspect"].value_counts())

elif menu == "Browse Reviews":
    st.title("рц"Ќ Browse Reviews")

    keyword = st.text_input("Cari Review")

    if keyword:
        filtered = df[df["text"].str.contains(keyword, case=False)]
    else:
        filtered = df

    st.dataframe(filtered[["text", "aspect", "sentiment"]])

elif menu == "Statistik":
    st.title("Statistik")

    st.subheader("Distribusi Sentimen")
    st.bar_chart(df["sentiment"].value_counts())

    st.subheader("Panjang Review")
    df["length"] = df["text"].apply(len)
    st.bar_chart(df["length"])

# ========================
elif menu == "Panel ABSA":
    st.title("Panel ABSA")

    aspek = st.selectbox("Pilih Aspek", df["aspect"].unique())
    sentimen = st.selectbox("Pilih Sentimen", df["sentiment"].unique())

    filtered = df[
        (df["aspect"] == aspek) &
        (df["sentiment"] == sentimen)
    ]

    st.write("Jumlah:", len(filtered))
    st.dataframe(filtered)
```

```python
import json
import pandas as pd
import matplotlib.pyplot as plt
from collections import Counter
from itertools import combinations

file_path = "../dataset/Kelp3_dataset_anotasi.jsonl"

data = []

with open(file_path, "r", encoding="utf-8") as f:
    for line in f:
        if line.strip():
            data.append(json.loads(line))

df = pd.DataFrame(data)

df.head()
```


```
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tapi baru sempat mampir. Beli baju katun lurik di sini. Suasana toko
syahdu. Staf toko semua informatif dan helpful. Pembayaran non tunai
bisa pakai QRIS.","[{'text': 'Sering', 'start': 0, 'end': 6, 'id': 0,
'ws': True}, {'text': 'banget', 'start': 7, 'end': 13, 'id': 1, 'ws':
True}, {'text': 'lewat', 'start': 14, 'end': 19, 'id': 2, 'ws': True},
{'text': 'tapi', 'start': 20, 'end': 24, 'id': 3, 'ws': True},
{'text': 'baru', 'start': 25, 'end': 29, 'id': 4, 'ws': True},
{'text': 'sempat', 'start': 30, 'end': 36, 'id': 5, 'ws': True},
{'text': 'mampir', 'start': 37, 'end': 43, 'id': 6, 'ws': False},

{'text': '.', 'start': 43, 'end': 44, 'id': 7, 'ws': True}, {'text': 'Beli', 'start': 45, 'end': 49, 'id': 8, 'ws': True}, {'text': 'baju', 'start': 50, 'end': 54, 'id': 9, 'ws': True}, {'text': 'katun', 'start': 55, 'end': 60, 'id': 10, 'ws': True}, {'text': 'lurik', 'start': 61, 'end': 66, 'id': 11, 'ws': True}, {'text': 'di', 'start': 67, 'end': 69, 'id': 12, 'ws': True}, {'text': 'sini', 'start': 70, 'end': 74, 'id': 13, 'ws': False}, {'text': '.', 'start': 74, 'end': 75, 'id': 14, 'ws': True}, {'text': 'Suasana', 'start': 76, 'end': 83, 'id': 15, 'ws': True}, {'text': 'toko', 'start': 84, 'end': 88, 'id': 16, 'ws': True}, {'text': 'syahdu', 'start': 89, 'end': 95, 'id': 17, 'ws': False}, {'text': '.', 'start': 95, 'end': 96, 'id': 18, 'ws': True}, {'text': 'Staf', 'start': 97, 'end': 101, 'id': 19, 'ws': True}, {'text': 'toko', 'start': 102, 'end': 106, 'id': 20, 'ws': True}, {'text': 'semua', 'start': 107, 'end': 112, 'id': 21, 'ws': True}, {'text': 'informatif', 'start': 113, 'end': 123, 'id': 22, 'ws': True}, {'text': 'dan', 'start': 124, 'end': 127, 'id': 23, 'ws': True}, {'text': 'helpful', 'start': 128, 'end': 135, 'id': 24, 'ws': False}, {'text': '.', 'start': 135, 'end': 136, 'id': 25, 'ws': True}, {'text': 'Pembayaran', 'start': 137, 'end': 147, 'id': 26, 'ws': True}, {'text': 'non', 'start': 148, 'end': 151, 'id': 27, 'ws': True}, {'text': 'tunai', 'start': 152, 'end': 157, 'id': 28, 'ws': True}, {'text': 'bisa', 'start': 158, 'end': 162, 'id': 29, 'ws': True}, {'text': 'pakai', 'start': 163, 'end': 168, 'id': 30, 'ws': True}, {'text': 'QRIS', 'start': 169, 'end': 173, 'id': 31, 'ws': False}, {'text': '.', 'start': 173, 'end': 174, 'id': 32, 'ws': False}]","[{'start': 50, 'end': 60, 'token_start': 9, 'token_end': 10, 'label': 'PRODUCT_POSITIVE'}, {'start': 76, 'end': 95, 'token_start': 15, 'token_end': 17, 'label': 'PLACE_POSITIVE'}, {'start': 97, 'end': 123, 'token_start': 19, 'token_end': 22, 'label': 'PLACE_POSITIVE'}, {'start': 137, 'end': 157, 'token_start': 26, 'token_end': 28, 'label': 'PROMOTION_NEUTRAL'}, {'start': 169, 'end': 173, 'token_start': 31, 'token_end': 31, 'label': 'PROMOTION_NEUTRAL'}]","-1365596721","1834013482","[{'id': 'PRODUCT_POSITIVE', 'text': 'PRODUCT_POSITIVE'}, {'id': 'PRODUCT_NEGATIVE', 'text': 'PRODUCT_NEGATIVE'}, {'id': 'PRODUCT_NEUTRAL', 'text': 'PRODUCT_NEUTRAL'}, {'id': 'PRICE_POSITIVE', 'text': 'PRICE_POSITIVE'}, {'id': 'PRICE_NEGATIVE', 'text': 'PRICE_NEGATIVE'}, {'id': 'PRICE_NEUTRAL', 'text': 'PRICE_NEUTRAL'}, {'id': 'PLACE_POSITIVE', 'text': 'PLACE_POSITIVE'}, {'id': 'PLACE_NEGATIVE', 'text': 'PLACE_NEGATIVE'}, {'id': 'PLACE_NEUTRAL', 'text': 'PLACE_NEUTRAL'}, {'id': 'PROMOTION_POSITIVE', 'text': 'PROMOTION_POSITIVE'}, {'id': 'PROMOTION_NEGATIVE', 'text': 'PROMOTION_NEGATIVE'}, {'id': 'PROMOTION_NEUTRAL', 'text': 'PROMOTION_NEUTRAL'}, {'id': 'OUT_OF_TOPIC', 'text': 'OUT_OF_TOPIC'}]","blocks","{'choice_style': 'multiple'}","['PRODUCT_POSITIVE', 'PLACE_POSITIVE', 'PROMOTION_NEUTRAL']","accept","1776736890","nlp3_genap2526-230712537","nlp3_genap2526-230712537"]],"shape": {"columns":13,"rows":5}}

```python
print("Jumlah data:", len(df))
print("Kolom dataset:", df.columns.tolist())

df[["text", "accept"]].head()
```

```
Jumlah data: 3169
Kolom dataset: ['text', 'tokens', 'spans', '_input_hash',
'_task_hash', 'options', '_view_id', 'config', 'accept', 'answer',
'_timestamp', '_annotator_id', '_session_id']
```

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"text","rawType":"object","type":"string"},
{"name":"accept","rawType":"object","type":"unknown"}],"ref":"95511545
-7061-4ca4-9dbe-b1616f658502","rows":[["0","Pusat penjualan benda-
benda seni dan segala kebutuhannya di kota Yogyakarta, sudah ada
sebelum Indonesia merdeka (1930an). Kebetulan saya membeli Tosan Aji
(senjata tradisional Jawa) yaitu keris. Toko ini mempunyai banyak
koleksi keris ...","['PRODUCT_POSITIVE']"],["1","Sering banget lewat
tapi baru sempat mampir. Beli baju katun lurik di sini. Suasana toko
syahdu. Staf toko semua informatif dan helpful. Pembayaran non tunai
bisa pakai QRIS.","['PRODUCT_POSITIVE', 'PLACE_POSITIVE',
'PROMOTION_NEUTRAL']"],["2","pelayanan nya ramah-ramah banget, barang
lengkap dari pakaian sampai cinderamata dan jntuk harga relatif lebih
murah dibanding toko sekelasnya","['PLACE_POSITIVE',
'PRODUCT_POSITIVE', 'PRICE_POSITIVE']"],["3","Pusat penjualan benda-
benda seni dan segala kebutuhannya di kota Yogyakarta, sudah ada
sebelum Indonesia merdeka (1930an). Kebetulan saya membeli Tosan Aji
(senjata tradisional Jawa) yaitu keris. Toko ini mempunyai banyak
koleksi keris ...","['PRODUCT_POSITIVE']"],["4","Sering banget lewat
tapi baru sempat mampir. Beli baju katun lurik di sini. Suasana toko
syahdu. Staf toko semua informatif dan helpful. Pembayaran non tunai
bisa pakai QRIS.","['PRODUCT_POSITIVE', 'PLACE_POSITIVE',
'PROMOTION_NEUTRAL']"]],"shape":{"columns":2,"rows":5}}

```python
df["accept"] = df["accept"].apply(lambda x: x if isinstance(x, list)
else [])

# Hitung jumlah label per review
df["num_labels"] = df["accept"].apply(len)

# Hitung panjang review berdasarkan jumlah kata
df["word_count"] = df["text"].apply(lambda x: len(str(x).split()))

df[["text", "accept", "num_labels", "word_count"]].head()
```

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"text","rawType":"object","type":"string"},
{"name":"accept","rawType":"object","type":"unknown"},
{"name":"num_labels","rawType":"int64","type":"integer"},
{"name":"word_count","rawType":"int64","type":"integer"}],"ref":"bc9f7
aa5-950f-4f98-a0ed-540bea994081","rows":[["0","Pusat penjualan benda-

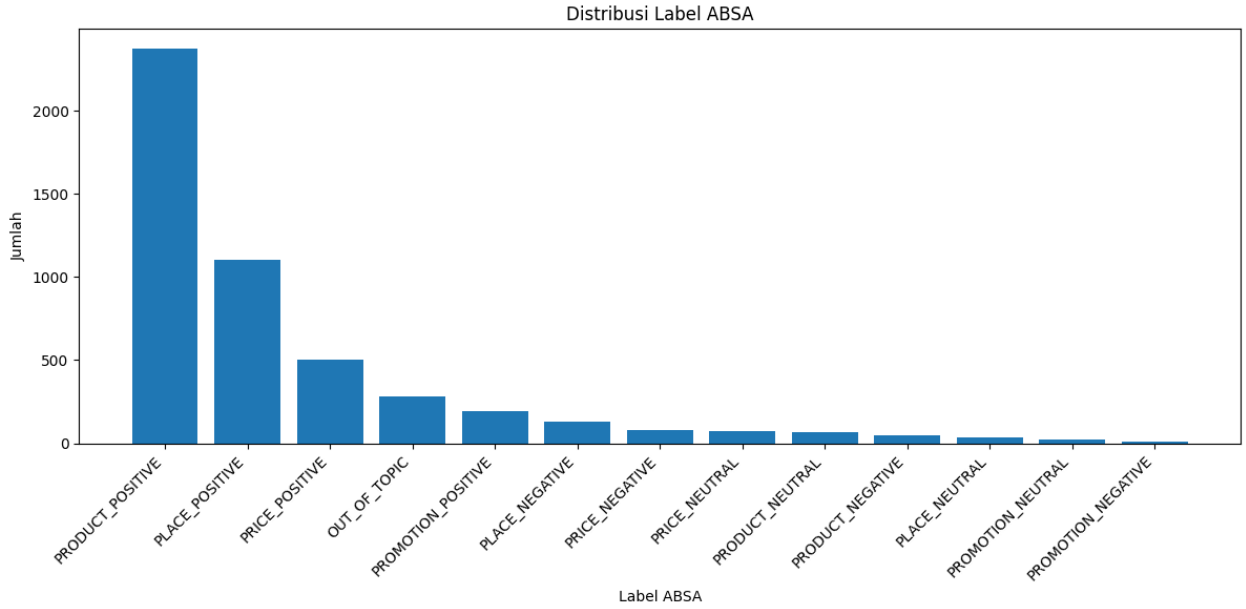
benda seni dan segala kebutuhannya di kota Yogyakarta, sudah ada sebelum Indonesia merdeka (1930an). Kebetulan saya membeli Tosan Aji (senjata tradisional Jawa) yaitu keris. Toko ini mempunyai banyak koleksi keris ...","['PRODUCT_POSITIVE']","1","33"],["1","Sering banget lewat tapi baru sempat mampir. Beli baju katun lurik di sini. Suasana toko syahdu. Staf toko semua informatif dan helpful. Pembayaran non tunai bisa pakai QRIS.","['PRODUCT_POSITIVE', 'PLACE_POSITIVE', 'PROMOTION_NEUTRAL']","3","28"],["2","pelayanan nya ramah-ramah banget, barang lengkap dari pakaian sampai cinderamata dan jntuk harga relatif lebih murah dibanding toko sekelasnya","['PLACE_POSITIVE', 'PRODUCT_POSITIVE', 'PRICE_POSITIVE']","3","19"],["3","Pusat penjualan benda-benda seni dan segala kebutuhannya di kota Yogyakarta, sudah ada sebelum Indonesia merdeka (1930an). Kebetulan saya membeli Tosan Aji (senjata tradisional Jawa) yaitu keris. Toko ini mempunyai banyak koleksi keris ...","['PRODUCT_POSITIVE']","1","33"],["4","Sering banget lewat tapi baru sempat mampir. Beli baju katun lurik di sini. Suasana toko syahdu. Staf toko semua informatif dan helpful. Pembayaran non tunai bisa pakai QRIS.","['PRODUCT_POSITIVE', 'PLACE_POSITIVE', 'PROMOTION_NEUTRAL']","3","28"]],"shape":{"columns":4,"rows":5}}

```
all_labels = []

for labels in df["accept"]:
    all_labels.extend(labels)

label_counts = Counter(all_labels)

label_df = pd.DataFrame(
    label_counts.items(),
    columns=["label", "count"]
).sort_values(by="count", ascending=False)

label_df
```

{"columns":[{"name":"index","rawType":"int64","type":"integer"},{"name":"label","rawType":"object","type":"string"},{"name":"count","rawType":"int64","type":"integer"}],"ref":"af759b19-986d-4c97-9d12-99e8bb899705","rows":[["0","PRODUCT_POSITIVE","2372"],["1","PLACE_POSITIVE","1106"],["3","PRICE_POSITIVE","503"],["5","OUT_OF_TOPIC","280"],["7","PROMOTION_POSITIVE","193"],["9","PLACE_NEGATIVE","128"],["4","PRICE_NEGATIVE","77"],["6","PRICE_NEUTRAL","73"],["10","PRODUCT_NEUTRAL","69"],["8","PRODUCT_NEGATIVE","50"],["11","PLACE_NEUTRAL","33"],["2","PROMOTION_NEUTRAL","24"],["12","PROMOTION_NEGATIVE","8"]],"shape":{"columns":2,"rows":13}}

```
plt.figure(figsize=(12, 6))
plt.bar(label_df["label"], label_df["count"])
plt.xticks(rotation=45, ha="right")
```

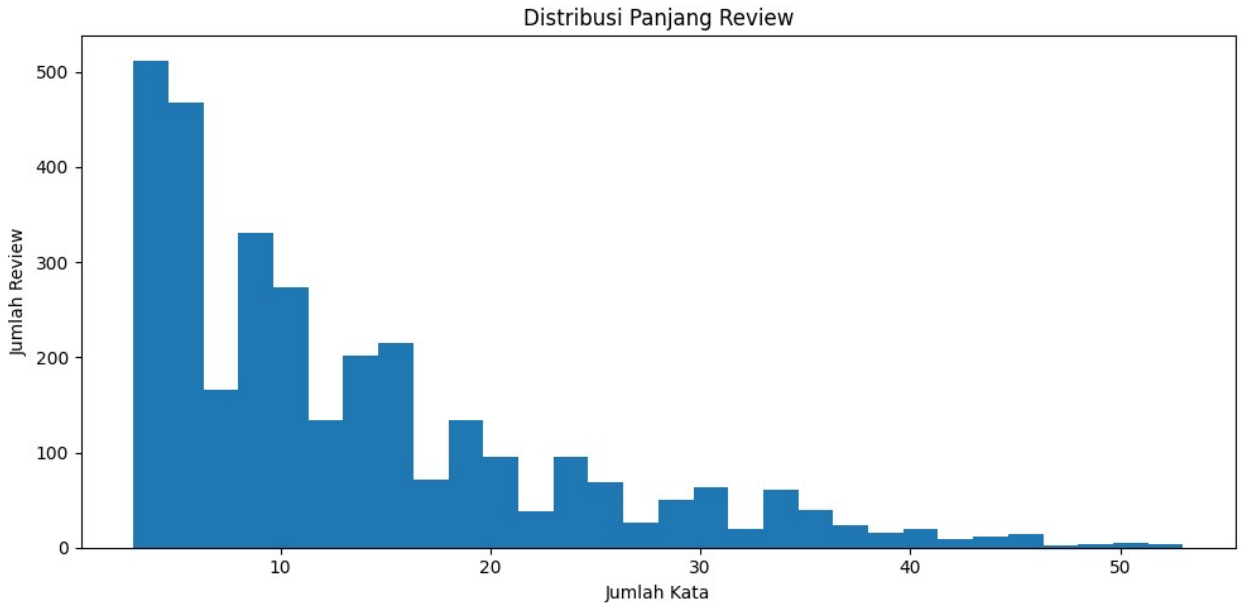
```
plt.xlabel("Label ABSA")
plt.ylabel("Jumlah")
plt.title("Distribusi Label ABSA")
plt.tight_layout()
plt.show()
```




```
df["word_count"].describe()
```

{"columns":[{"name":"index","rawType":"object","type":"string"},{"name":"word_count","rawType":"float64","type":"float"}],"ref":"33f06da2-1cfc-43df-9101-0da70a1c7236","rows":[["count","3169.0"],["mean","13.209845377090565"],["std","9.82002587005034"],["min","3.0"],["25%","6.0"],["50%","10.0"],["75%","18.0"],["max","53.0"]],"shape":{"columns":1,"rows":8}}

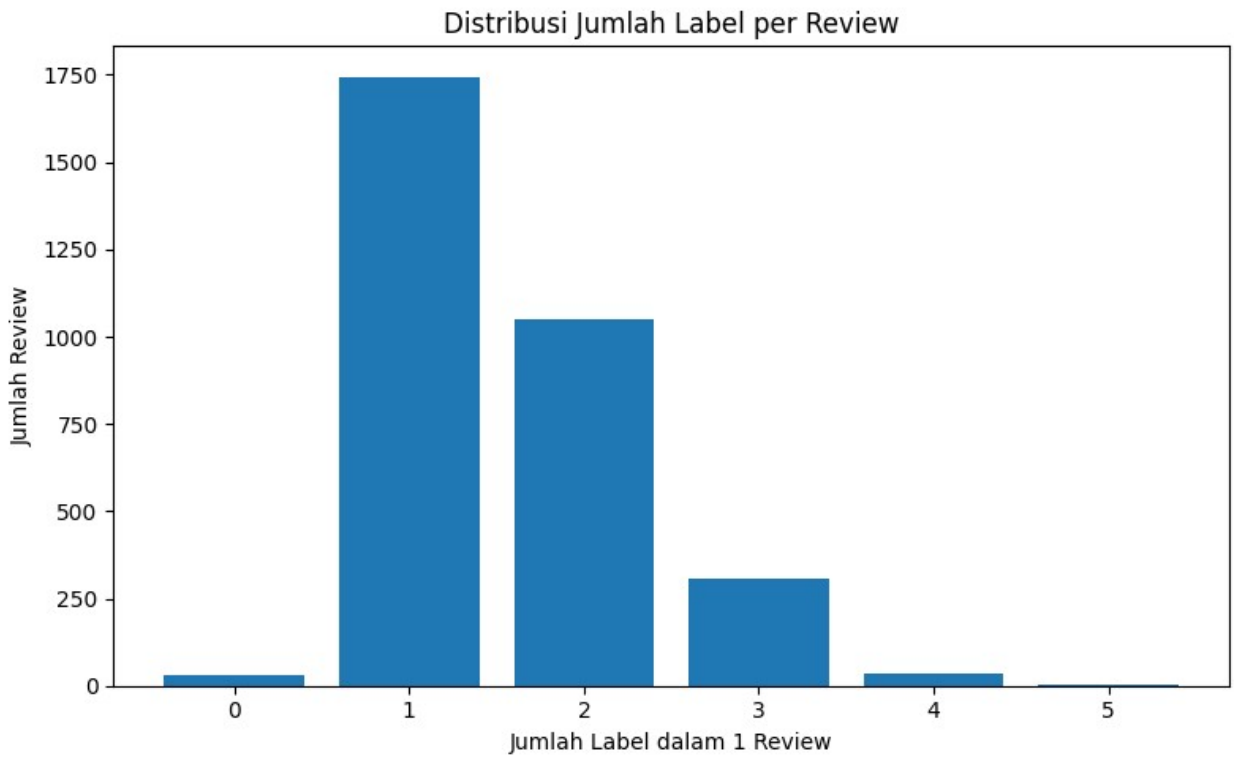
```
plt.figure(figsize=(10, 5))
plt.hist(df["word_count"], bins=30)
plt.xlabel("Jumlah Kata")
plt.ylabel("Jumlah Review")
plt.title("Distribusi Panjang Review")
plt.tight_layout()
plt.show()
```




```
num_label_counts = df["num_labels"].value_counts().sort_index()

num_label_counts
```

{"columns":[{"name":"num_labels","rawType":"int64","type":"integer"},{"name":"count","rawType":"int64","type":"integer"}],"ref":"160ce5ff-6be9-4361-af26-8759adc54075","rows":[["0","31"],["1","1745"],["2","1049"],["3","306"],["4","35"],["5","3"]],"shape":{"columns":1,"rows":6}}

```
plt.figure(figsize=(8, 5))
plt.bar(num_label_counts.index.astype(str), num_label_counts.values)
plt.xlabel("Jumlah Label dalam 1 Review")
plt.ylabel("Jumlah Review")
plt.title("Distribusi Jumlah Label per Review")
plt.tight_layout()
plt.show()
```




```
all_unique_labels = sorted(set(all_labels))

for label in all_unique_labels:
    df[label] = df["accept"].apply(lambda labels: 1 if label in labels
else 0)

df[["text"] + all_unique_labels].head()
```

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"text","rawType":"object","type":"string"},
{"name":"OUT_OF_TOPIC","rawType":"int64","type":"integer"},
{"name":"PLACE_NEGATIVE","rawType":"int64","type":"integer"},
{"name":"PLACE_NEUTRAL","rawType":"int64","type":"integer"},
{"name":"PLACE_POSITIVE","rawType":"int64","type":"integer"},
{"name":"PRICE_NEGATIVE","rawType":"int64","type":"integer"},
{"name":"PRICE_NEUTRAL","rawType":"int64","type":"integer"},
{"name":"PRICE_POSITIVE","rawType":"int64","type":"integer"},
{"name":"PRODUCT_NEGATIVE","rawType":"int64","type":"integer"},
{"name":"PRODUCT_NEUTRAL","rawType":"int64","type":"integer"},
{"name":"PRODUCT_POSITIVE","rawType":"int64","type":"integer"},
{"name":"PROMOTION_NEGATIVE","rawType":"int64","type":"integer"},
{"name":"PROMOTION_NEUTRAL","rawType":"int64","type":"integer"},
{"name":"PROMOTION_POSITIVE","rawType":"int64","type":"integer"}],"ref
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benda-benda seni dan segala kebutuhannya di kota Yogyakarta, sudah ada
sebelum Indonesia merdeka (1930an). Kebetulan saya membeli Tosan Aji

(senjata tradisional Jawa) yaitu keris. Toko ini mempunyai banyak koleksi
keris ...","0","0","0","0","0","0","0","0","0","1","0","0","0"],
["1","Sering banget lewat tapi baru sempat mampir. Beli baju katun lurik di sini. Suasana toko syahdu. Staf toko semua informatif dan helpful. Pembayaran non tunai bisa pakai
QRIS.","0","0","0","1","0","0","0","0","0","1","0","1","0"],
["2","pelayanan nya ramah-ramah banget, barang lengkap dari pakaian sampai cinderamata dan jntuk harga relatif lebih murah dibanding toko sekelasnya","0","0","0","1","0","0","1","0","0","1","0","0","0"],
["3","Pusat penjualan benda-benda seni dan segala kebutuhannya di kota Yogyakarta, sudah ada sebelum Indonesia merdeka (1930an). Kebetulan saya membeli Tosan Aji (senjata tradisional Jawa) yaitu keris. Toko ini mempunyai banyak koleksi
keris ...","0","0","0","0","0","0","0","0","0","1","0","0","0"],
["4","Sering banget lewat tapi baru sempat mampir. Beli baju katun lurik di sini. Suasana toko syahdu. Staf toko semua informatif dan helpful. Pembayaran non tunai bisa pakai
QRIS.","0","0","0","1","0","0","0","0","0","1","0","1","0"]],"shape":
{"columns":14,"rows":5}}

```
corr_matrix = df[all_unique_labels].corr()
```

```
corr_matrix
```

{"columns":[{"name":"index","rawType":"object","type":"string"},
{"name":"OUT_OF_TOPIC","rawType":"float64","type":"float"},
{"name":"PLACE_NEGATIVE","rawType":"float64","type":"float"},
{"name":"PLACE_NEUTRAL","rawType":"float64","type":"float"},
{"name":"PLACE_POSITIVE","rawType":"float64","type":"float"},
{"name":"PRICE_NEGATIVE","rawType":"float64","type":"float"},
{"name":"PRICE_NEUTRAL","rawType":"float64","type":"float"},
{"name":"PRICE_POSITIVE","rawType":"float64","type":"float"},
{"name":"PRODUCT_NEGATIVE","rawType":"float64","type":"float"},
{"name":"PRODUCT_NEUTRAL","rawType":"float64","type":"float"},
{"name":"PRODUCT_POSITIVE","rawType":"float64","type":"float"},
{"name":"PROMOTION_NEGATIVE","rawType":"float64","type":"float"},
{"name":"PROMOTION_NEUTRAL","rawType":"float64","type":"float"},
{"name":"PROMOTION_POSITIVE","rawType":"float64","type":"float"}],"ref
":"7e3fdc98-2b73-457a-b8ee-a59f02e413c3","rows":
[["OUT_OF_TOPIC","1.0","-0.06387075053319666","-
0.020982752082155307","-0.19062486742443455","-0.04912820139474017","-
0.04039269139669809","-0.10784161337043043","-0.030494609734614993","-
0.046446066879734633","-0.4627573676586831","-0.015661667166255604","-
0.027195718720736405","-0.05603490110990742"],["PLACE_NEGATIVE","-
0.06387075053319666","1.0","0.026322214195289476","-
0.06279001040379846","0.08213226897952161","-0.02081909567304298","-
0.005775919406679695","0.19268184352505907","-0.008643265664992681","-
0.12489465048450211","0.02162005827434493","0.0005659037784021361","-
0.05224671215381564"],["PLACE_NEUTRAL","-

0.020982752082155307","0.026322214195289476","1.0","-
0.029459079873536322","0.004000824873366577","0.0671330933500442","-
0.027544229330110012","0.01195694103006404","0.06989420814415541","-
0.01934820954144231","-0.00516061771073992","-0.00896116014959901","-
0.0001271515266590403 2"],["PLACE_POSITIVE","-0.19062486742443455","-
0.06279001040379846","-0.029459079873536322","1.0","-
0.07254945353295261","-0.015345905199269793","0.04973053092632319","-
0.06614273982614337","-0.07294808471159075","-0.0501119248507246","-
0.03683500177121047","-0.0028723220499681213","-0.0369782617305406"],
["PRICE_NEGATIVE","-
0.04912820139474017","0.08213226897952161","0.004000824873366577","-
0.07254945353295261","1.0","0.0030909511710002853","-
0.03489487654448584","0.09514275955122614","-0.023543407432904114","-
0.03133894673235505","0.03290246745184483","-0.013785449001114855","-
0.03161757690118391"],["PRICE_NEUTRAL","-0.04039269139669809","-
0.02081909567304298","0.0671330933500442","-
0.015345905199269793","0.0030909511710002853","1.0","-
0.032160290273262726","-
0.0025620669886350102","0.020330133232791595","0.030831261569946554","
-
0.0077249148257769506","0.010849057186990061","0.022464987229502232"],
["PRICE_POSITIVE","-0.10784161337043043","-0.005775919406679695","-
0.027544229330110012","0.04973053092632319","-0.03489487654448584","-
0.032160290273262726","1.0","-0.04113658311048583","-
0.041135307971467375","0.1642087833979381","0.029774668807903876","0.0
11859238259230303","0.03381966846753602"],["PRODUCT_NEGATIVE","-
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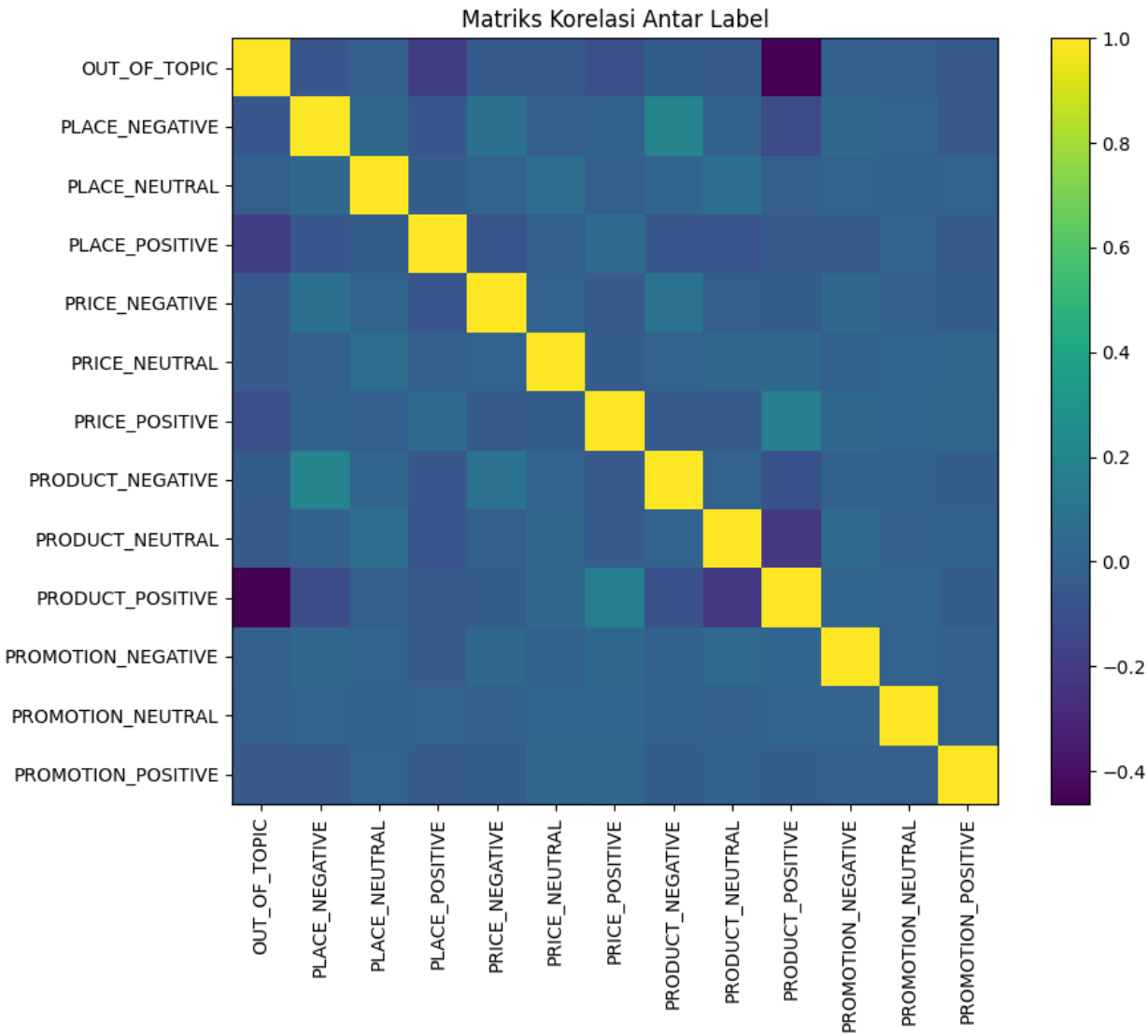
0.06614273982614337","0.09514275955122614","-0.0025620669886350102","-
0.04113658311048583","1.0","-0.0015385380383176204","-
0.09586275035231068","-0.006369569202169907","-
0.011060445261389943","-0.0322433016169334"],["PRODUCT_NEUTRAL","-
0.046446066879734633","-0.008643265664992681","0.06989420814415541","-
0.07294808471159075","-0.023543407432904114","0.020330133232791595","-
0.041135307971467375","-0.0015385380383176204","1.0","-
0.21751064970305792","0.03558287942059239","-0.013032837923952642","-
0.010869872887023893"],["PRODUCT_POSITIVE","-0.4627573676586831","-
0.12489465048450211","-0.01934820954144231","-0.0501119248507246","-
0.03133894673235505","0.030831261569946554","0.1642087833979381","-
0.09586275035231068","-
0.21751064970305792","1.0","0.014667470231690249","0.00030178830270911
86","-0.028771631771534998"],["PROMOTION_NEGATIVE","-
0.015661667166255604","0.02162005827434493","-0.00516061771073992","-
0.03683500177121047","0.03290246745184483","-
0.0077249148257769506","0.029774668807903876","-
0.006369569202169907","0.03558287942059239","0.014667470231690249","1.
0","-0.004394687936122308 6","-0.012811351196803697"],
["PROMOTION_NEUTRAL","-
0.027195718720736405","0.0005659037784021361","-
0.00896116014959901","-0.0028723220499681213","-

0.013785449001114855","0.010849057186990061","0.011859238259230303","-0.011060445261389943","-0.013032837923952642","0.0003017883027091186","-0.0043946879361223086","1.0","-0.022246284503576436"],["PROMOTION_POSITIVE","-0.05603490110990742","-0.05224671215381564","-0.00012715152665904032","-0.0369782617305406","-0.03161757690118391","0.022464987229502232","0.03381966846753602","-0.0322433016169334","-0.010869872887023893","-0.028771631771534998","-0.012811351196803697","-0.022246284503576436","1.0"]],"shape":{"columns":13,"rows":13}}

```
plt.figure(figsize=(10, 8))
plt.imshow(corr_matrix)
plt.xticks(range(len(all_unique_labels)), all_unique_labels,
rotation=90)
plt.yticks(range(len(all_unique_labels)), all_unique_labels)
plt.colorbar()
plt.title("Matriks Korelasi Antar Label")
plt.tight_layout()
plt.show()
```

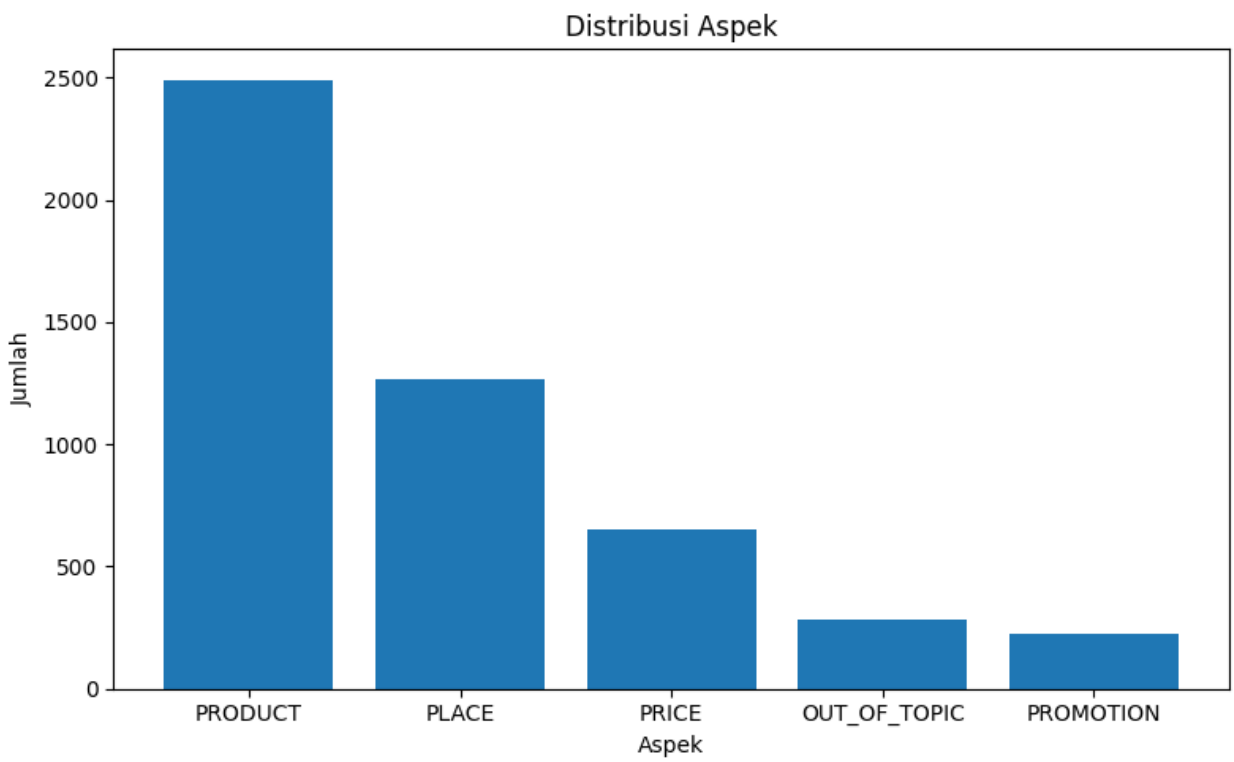



```
def get_aspect(label):
    if label == "OUT_OF_TOPIC":
        return "OUT_OF_TOPIC"
    return label.split("_")[0]

aspect_counts = Counter([get_aspect(label) for label in all_labels])

aspect_df = pd.DataFrame(
    aspect_counts.items(),
    columns=["aspect", "count"]
).sort_values(by="count", ascending=False)

aspect_df
```

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"aspect","rawType":"object","type":"string"},

{"name":"count","rawType":"int64","type":"integer"}],"ref":"92f8cdca-d86a-40e8-a798-1dbfd86ed278","rows":[["0","PRODUCT","2491"],["1","PLACE","1267"],["3","PRICE","653"],["4","OUT_OF_TOPIC","280"],["2","PROMOTION","225"]],"shape":{"columns":2,"rows":5}}

```
plt.figure(figsize=(8, 5))
plt.bar(aspect_df["aspect"], aspect_df["count"])
plt.xlabel("Aspek")
plt.ylabel("Jumlah")
plt.title("Distribusi Aspek")
plt.tight_layout()
plt.show()
```

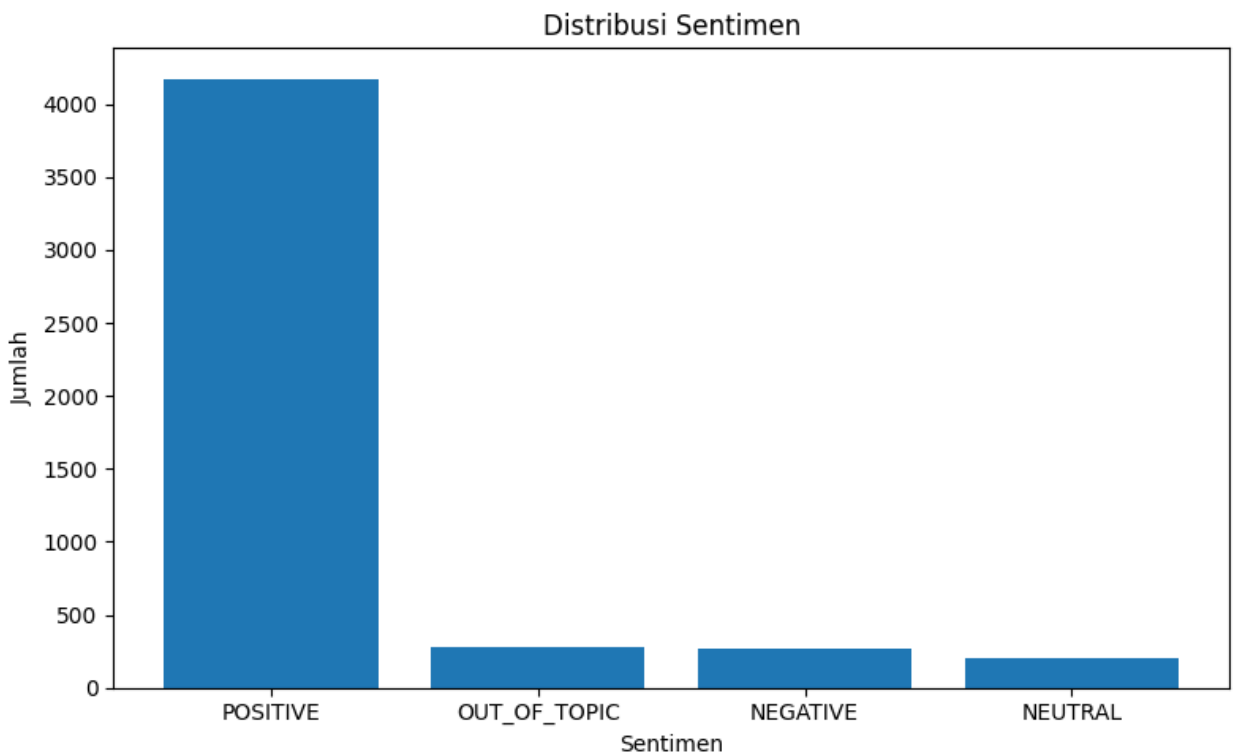



```
def get_sentiment(label):
    if label == "OUT_OF_TOPIC":
        return "OUT_OF_TOPIC"
    return label.split("_")[-1]

sentiment_counts = Counter([get_sentiment(label) for label in
all_labels])

sentiment_df = pd.DataFrame(
    sentiment_counts.items(),
    columns=["sentiment", "count"]
).sort_values(by="count", ascending=False)
```

```
sentiment_df
```

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"sentiment","rawType":"object","type":"string"},
{"name":"count","rawType":"int64","type":"integer"}],"ref":"3880fa03-76d0-47df-9f58-b7b608d8514b","rows":[["0","POSITIVE","4174"],
["3","OUT_OF_TOPIC","280"],["2","NEGATIVE","263"],
["1","NEUTRAL","199"]],"shape":{"columns":2,"rows":4}}

```
plt.figure(figsize=(8, 5))
plt.bar(sentiment_df["sentiment"], sentiment_df["count"])
plt.xlabel("Sentimen")
plt.ylabel("Jumlah")
plt.title("Distribusi Sentimen")
plt.tight_layout()
plt.show()
```

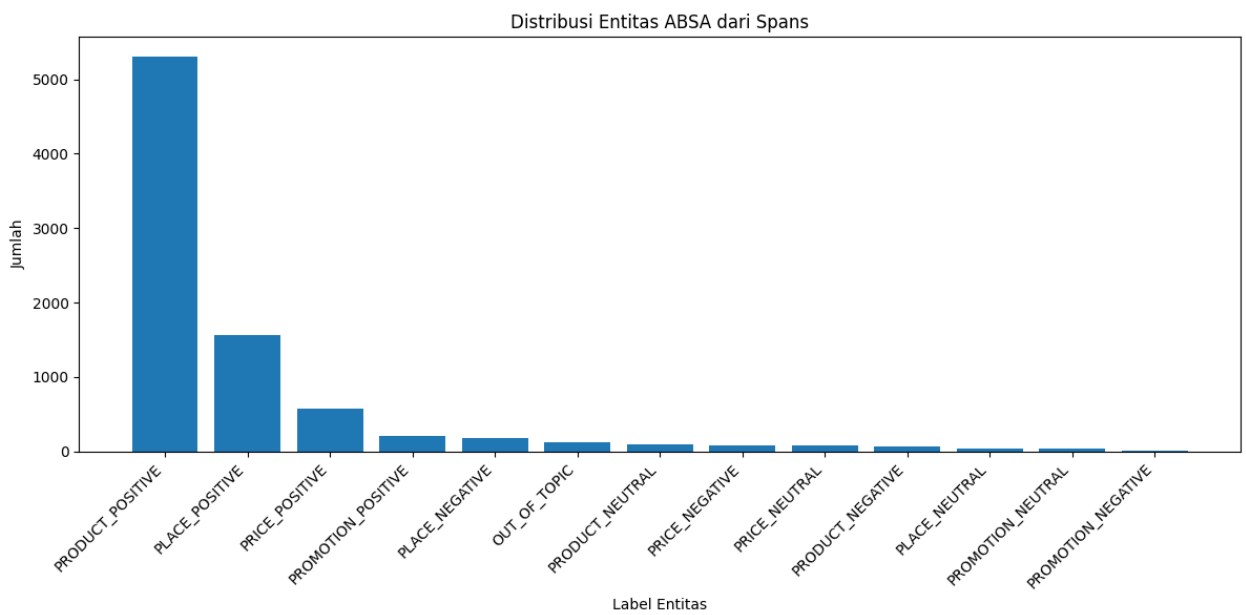



```
all_span_labels = []

if "spans" in df.columns:
    for spans in df["spans"]:
        if isinstance(spans, list):
            for span in spans:
                all_span_labels.append(span.get("label"))
```

```
span_counts = Counter(all_span_labels)

span_df = pd.DataFrame(
    span_counts.items(),
    columns=["entity_label", "count"]
).sort_values(by="count", ascending=False)

span_df
```

{"columns":[{"name":"index","rawType":"int64","type":"integer"},
{"name":"entity_label","rawType":"object","type":"string"},
{"name":"count","rawType":"int64","type":"integer"}],"ref":"f354b426-5a66-4df4-8a79-e94edc1608eb","rows":[["1","PRODUCT_POSITIVE","5304"],
["2","PLACE_POSITIVE","1560"],["4","PRICE_POSITIVE","570"],
["7","PROMOTION_POSITIVE","207"],["11","PLACE_NEGATIVE","184"],
["0","OUT_OF_TOPIC","125"],["9","PRODUCT_NEUTRAL","90"],
["5","PRICE_NEGATIVE","85"],["6","PRICE_NEUTRAL","74"],
["8","PRODUCT_NEGATIVE","62"],["10","PLACE_NEUTRAL","43"],
["3","PROMOTION_NEUTRAL","31"],
["12","PROMOTION_NEGATIVE","7"]],"shape":{"columns":2,"rows":13}}

```
plt.figure(figsize=(12, 6))
plt.bar(span_df["entity_label"], span_df["count"])
plt.xticks(rotation=45, ha="right")
plt.xlabel("Label Entitas")
plt.ylabel("Jumlah")
plt.title("Distribusi Entitas ABSA dari Spans")
plt.tight_layout()
plt.show()
```



```python
print("Jumlah total review:", len(df))
print("Rata-rata panjang review:", round(df["word_count"].mean(), 2),
"kata")
print("Review terpendek:", df["word_count"].min(), "kata")
print("Review terpanjang:", df["word_count"].max(), "kata")
print("Rata-rata jumlah label per review:",
round(df["num_labels"].mean(), 2))

print("\nLabel paling dominan:")
print(label_df.head(5))

print("\nAspek paling dominan:")
print(aspect_df.head())

print("\nSentimen paling dominan:")
print(sentiment_df.head())
```

```
Jumlah total review: 3169
Rata-rata panjang review: 13.21 kata
Review terpendek: 3 kata
Review terpanjang: 53 kata
Rata-rata jumlah label per review: 1.55

Label paling dominan:
                 label  count
0    PRODUCT_POSITIVE   2372
1      PLACE_POSITIVE   1106
3      PRICE_POSITIVE    503
5        OUT_OF_TOPIC    280
7  PROMOTION_POSITIVE    193

Aspek paling dominan:
         aspect  count
0       PRODUCT   2491
1         PLACE   1267
3         PRICE    653
4  OUT_OF_TOPIC    280
2     PROMOTION    225

Sentimen paling dominan:
      sentiment  count
0      POSITIVE   4174
3  OUT_OF_TOPIC    280
2      NEGATIVE    263
1       NEUTRAL    199
```